PHYS1408 Practice Exam 2-5 — Answer Key

Worked solutions

All answers were checked numerically with `_scripts/build_exam_2_5.py`. $g = 9.8$ m/s^2 throughout.

Practice Exam 2-5 reuses the exam's problem types with new twists: a coaster with a starting speed, the compression **needed** for a launch, N at the **bottom** of a loop, a rebound at a different speed, a collision where the carts stop dead, a child landing **inside** the rim. Question 1 is a **minimum** trap: round up to the choice that works. The last column says which 3×5 notecard entry solves each question.

Q	Answer	Value	Topic	Notecard entry
1	**c**	18.2 m/s → 19	Banked curve, minimum speed	Curve
2	**b**	2.95 m/s	Atwood's machine, disk pulley	Atwood / mass on pulley
3	**c**	20.4 m/s	Roller coaster with a starting speed	Fall / coaster
4	**a**	4.13 m/s	Sliding up a rough incline	One block on incline (Up)
5	**e**	0.11 m	Vertical spring launch, find the compression	Spring
6	**d**	3530 N	Loop, normal force at the bottom	Loop
7	**b**	3.01 m/s	Block down an incline with a disk pulley	Incline + hanging mass + pulley ($m_2 = 0$)
8	**e**	200 N	Force from a change in momentum	Force from Δp
9	**a**	0%	Inelastic collision, KE fraction	Stick (inelastic)
10	**d**	+3.2 m/s	1-D elastic collision, head-on	Elastic
11	**b**	0.5%	Ballistic pendulum, KE kept	Ballistic
12	**d**	0.35 kg · m^2	Parallel axis theorem, rod	Parallel axis
13	**e**	67 rad/s	Rolling sphere, find ω	Rolling (sphere row)

Q	Answer	Value	Topic	Notecard entry
14	**c**	0.98 rad/s	Person runs and jumps on inside the rim	Angular momentum
15	**a**	5.0 s	Rotational kinematics, time from an angle	Rot. kinematics

1. Banked curve, minimum speed — c) 19 m/s

Source: Ch 6 notes p. 7/7; handout p. 2/3. **Notecard:** Curve.

At v_{min} the car tends to slide **down** the bank, so friction points **up** the bank. The notecard formula with the $-$ signs:

$$v_{min} = \sqrt{gr\,\frac{\tan\theta - \mu_s}{1 + \mu_s \tan\theta}}$$

Plug in ($\tan 25^\circ = 0.4663$):

$$v_{min} = \sqrt{(9.8)(103)\,\frac{0.4663 - 0.12}{1 + (0.12)(0.4663)}}$$

$$= \sqrt{1009.4 \times \frac{0.3463}{1.0560}} = \sqrt{331.0} = 18.2 \text{ m/s}$$

Answer-choice trap. 18.2 m/s is a **minimum**. At 18 m/s the car slides down the bank, even though 18 is the closest number. The smallest choice that works is **c) 19 m/s**.

Distractors:

- a) 11 is the flat-curve formula, $\sqrt{\mu_s g r}$.
- b) 18 is the nearest number, but it's below the minimum.
- d) 22 is the frictionless bank, $\sqrt{gr\tan\theta}$.
- e) 25 is v_{max}.

2. Atwood's machine, disk pulley — b) 2.95 m/s

Source: Ch 7/8 notes p. 2/18 + Ch 10 notes p. 4/8. **Notecard:** Atwood / mass on pulley.

Notecard (two masses):

$$v_f = \sqrt{v_i^2 + \frac{2gh(m_1 - m_2)}{m_1 + m_2 + I/R^2}}$$

A disk has $I/R^2 = \frac{1}{2}M = 1.0$ kg (R cancels).

Plug in ($v_i = 0$, $h = 2.0$ m):

$$v_f = \sqrt{\frac{2(9.8)(2.0)(5.0 - 3.0)}{5.0 + 3.0 + 1.0}}$$

$$= \sqrt{\frac{78.4}{9.0}} = \sqrt{8.71} = 2.95 \text{ m/s}$$

Distractors:

- a) 2.80 treats the pulley as a ring ($I/R^2 = M$).
- c) 3.13 ignores the pulley's mass.
- d) 3.61 leaves m_2 out of the denominator.
- e) 6.26 is free fall, $\sqrt{2gh}$.

3. Roller coaster with a starting speed — c) 20.4 m/s

Source: Ch 7/8 notes p. 3/18. **Notecard:** Fall / coaster.

Notecard (no friction):

$$v_f = \sqrt{v_i^2 + 2g(h_i - h_f)}$$

Plug in:

$$v_f = \sqrt{8.0^2 + 2(9.8)(30 - 12)} = \sqrt{64 + 352.8} = \sqrt{416.8} = 20.4 \text{ m/s}$$

Add the KE you start with **inside** the root, not to the final speed.

Distractors:

- a) 15.5 forgets the 2 in $2g$.
- b) 18.8 ignores the starting speed.
- d) 25.5 uses the full 30 m drop.
- e) 26.8 adds 8.0 m/s to the speed from rest ($18.8 + 8.0$).

4. Sliding up a rough incline — a) 4.13 m/s

Source: Ch 7/8 notes p. 7/18–8/18. **Notecard:** One block on incline (Up).

Notecard, sliding **up**:

$$v_f = \sqrt{v_i^2 - 2g\,\Delta x(\sin\theta + \mu\cos\theta)}$$

Plug in ($\sin 20^\circ = 0.3420$, $\cos 20^\circ = 0.9397$):

$$v_f = \sqrt{6.0^2 - 2(9.8)(2.0)(0.3420 + 0.15 \times 0.9397)}$$

$$= \sqrt{36 - 39.2(0.4830)} = \sqrt{36 - 18.93} = \sqrt{17.07} = 4.13 \text{ m/s}$$

The number under the root is positive, so it does make it 2.0 m (e is wrong).

Distractors:

- b) 4.75 ignores friction.
- c) 5.15 forgets the 2 in $2g$.
- d) 7.41 adds instead of subtracts (that's the sliding-down sign).
- e) It doesn't stop: $v_f^2 = 17.07 > 0$.

5. Vertical spring launch, find the compression — e) 0.11 m

Source: Ch 7/8 notes p. 12/18; 2015 Exam 2 #5. **Notecard:** Spring.

Notecard launch line, $h = \dfrac{kx^2}{2mg}$. **Solve for x:**

$$kx^2 = 2mgh$$

$$x = \sqrt{\frac{2mgh}{k}}$$

Plug in:

$$x = \sqrt{\frac{2(0.25)(9.8)(3.0)}{1200}} = \sqrt{\frac{14.7}{1200}} = \sqrt{0.01225} = 0.11 \text{ m}$$

Distractors:

- a) 0.006 is mgh/k (no 2, no root).
- b) 0.012 forgets the square root.
- c) 0.055 is $\sqrt{mgh/2k}$ (the 2 upside down).
- d) 0.078 drops the 2: $\sqrt{mgh/k}$.

6. Loop, normal force at the bottom — d) 3530 N

Source: Ch 7/8 notes p. 13/18. **Notecard:** Loop.

Speed at the bottom (energy from the start, drop $h = 20$ m):

$$v_b^2 = 2gh = 2(9.8)(20) = 392 \text{ m}^2/\text{s}^2$$

Forces at the bottom. N points up, toward the center, and mg points down. Notecard:

$$N_{bot} = \frac{mv_b^2}{R} + mg$$

Plug in:

$$N_{bot} = \frac{(60)(392)}{8.0} + (60)(9.8) = 2940 + 588 = 3528 \approx 3530 \text{ N}$$

That's 6 times the rider's weight.

Distractors:

- a) 588 N is just the weight mg.
- b) 2350 subtracts mg (the top-of-loop sign).
- c) 2940 forgets to add mg.
- e) 7060 doubles it.

7. Block down an incline with a disk pulley — b) 3.01 m/s

Source: 2015 Exam 2 #13. **Notecard:** Incline + hanging mass + pulley ($m_2 = 0$).

Notecard, incline + pulley, with no hanging mass ($m_2 = 0$) and the block sliding **down**:

$$v_f = \sqrt{v_i^2 + \frac{2g\,\Delta x\,B}{m + I/R^2}}$$

$$B = m\sin\theta - \mu m\cos\theta$$

Net pull ($\sin 35^\circ = 0.5736$, $\cos 35^\circ = 0.8192$):

$$B = 3.0(0.5736) - 0.20(3.0)(0.8192) = 1.721 - 0.492 = 1.229 \text{ kg}$$

Pulley term: disk, $I/R^2 = \frac{1}{2}M = 1.0$ kg.

Plug in ($v_i = 0$):

$$v_f = \sqrt{\frac{2(9.8)(1.5)(1.229)}{3.0 + 1.0}}$$

$$= \sqrt{\frac{36.14}{4.0}} = \sqrt{9.04} = 3.01 \text{ m/s}$$

Distractors:

- a) 2.69 treats the pulley as a ring.
- c) 3.47 ignores the pulley's mass.
- d) 3.56 ignores friction.
- e) 4.11 ignores both friction and the pulley.

8. Force from a change in momentum — e) 200 N

Source: Ch 9 notes p. 1/5. **Notecard:** Force from Δp.

Notecard: $F = \dfrac{m(v_f - v_i)}{t}$ with **signs**. Take "away from the wall" as +, so $v_i = -15$ m/s and $v_f = +10$ m/s.

$$F = \frac{m(v_f - v_i)}{t}$$

Plug in:

$$F = \frac{0.40\,(10 - (-15))}{0.050} = \frac{0.40(25)}{0.050} = 200 \text{ N}$$

Distractors:

- a) 40 uses $15 - 10$ (ignores the reversal).
- b) 80 uses only the rebound speed.
- c) 120 uses only the incoming speed (as if it stopped).
- d) 160 uses $15 + 5$.

9. Inelastic collision, KE fraction — a) 0%

Source: Ch 9 notes p. 1/5; 2015 Exam 2 #7. **Notecard:** Stick (inelastic).

Symbolic masses and speeds, so **set** $m = 1$ **and** $v = 1$. Right is +: $m_1 = 1$, $v_1 = +1$; $m_2 = 3$, $v_2 = -\frac{1}{3}$.

Final velocity (notecard):

$$v_f = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \frac{(1)(1) + (3)(-\frac{1}{3})}{4} = \frac{1 - 1}{4} = 0$$

The momenta are equal and opposite, so the carts **stop dead**. Then

$$\frac{K_f}{K_i} = \frac{(m_1 + m_2)v_f^2}{m_1 v_1^2 + m_2 v_2^2} = \frac{4(0)^2}{1 + 3(\frac{1}{9})} = 0$$

All of the kinetic energy is lost.

Distractors:

- b) 25% is the mass ratio $\frac{m}{4m}$, which only applies when one cart starts at rest.
- c) and d) come from treating the second cart as moving the same way.
- e) 100% would need an elastic collision.

10. 1-D elastic collision, head-on — d) +3.2 m/s

Source: Ch 9 notes p. 2/5–3/5; 2015 Exam 2 #6. **Notecard:** Elastic.

Notecard formulas, right = +: $m_1 = 2.0$, $v_{1i} = +5.0$; $m_2 = 3.0$, $v_{2i} = -4.0$.

$$v_{1f} = \frac{(m_1 - m_2)v_{1i} + 2m_2 v_{2i}}{m_1 + m_2} = \frac{(-1.0)(5.0) + 2(3.0)(-4.0)}{5.0} = \frac{-5 - 24}{5} = -5.8 \text{ m/s}$$

$$v_{2f} = v_{1i} + v_{1f} - v_{2i} = 5.0 + (-5.8) - (-4.0) = +3.2 \text{ m/s}$$

The 3.0 kg ball reverses and moves right at 3.2 m/s.

Distractors:

- a) +5.8 is the size of the 2.0 kg ball's velocity, not the 3.0 kg ball's.
- b) +5.0 assumes the balls swap velocities.
- c) +4.0 assumes the 3.0 kg ball just reverses at the same speed.
- e) −0.4 is the stick-together answer.

11. Ballistic pendulum, KE kept — b) 0.1–1%

Source: Ch 9 notes p. 5/5; 2015 Exam 2 #8. **Notecard:** Ballistic.

Notecard: $\dfrac{K_f}{K_i} = \dfrac{m}{M + m}$ (no speed or height needed).

$$\frac{K_f}{K_i} = \frac{0.020}{3.98 + 0.020} = \frac{0.020}{4.00} = 0.005 = 0.5\%$$

That falls in the 0.1–1% range.

Distractors:

- a) $< 0.1\%$ would need a much heavier block (2015 Exam #8 used $M = 10$ kg and a 5 g bullet).
- e) 50% would mean the collision kept most of the KE; sticking collisions with $m \ll M$ lose almost all of it.

12. Parallel axis theorem, rod — d) 0.35 kg · m²

Source: Ch 10 notes p. 2/8. **Notecard:** Parallel axis.

Notecard rod shortcut, axis a from one end:

$$I = m\left[\frac{\ell^2}{12} + \left(\frac{\ell}{2} - a\right)^2\right]$$

$d = \dfrac{\ell}{2} - a = 0.50 - 0.20 = 0.30$ m from the **center**.

Plug in:

$$I = 2.0\left[\frac{1.0^2}{12} + (0.50 - 0.20)^2\right]$$

$$= 2.0(0.0833 + 0.090) = 0.35 \text{ kg} \cdot \text{m}^2$$

Check: it's between $I_{cm} = 0.17$ and $I_{end} = \frac{1}{3}m\ell^2 = 0.67$.

Distractors:

- a) 0.08 is ma^2, measured from the end.
- b) 0.17 is I_{cm} alone.
- c) 0.25 is the #1 mistake: $I_{cm} + ma^2$, using the distance from the **end**.
- e) 0.67 is the rod about its end.

13. Rolling sphere, find ω — e) 67 rad/s

Source: Ch 10 notes p. 6/8–8/8; 2015 Exam 2 #10. **Notecard:** Rolling (sphere row).

Notecard, sphere row: $v = \sqrt{\frac{10}{7}gh}$, then $\omega = \dfrac{v}{R}$.

Speed:

$$v = \sqrt{\tfrac{10}{7}(9.8)(0.80)} = \sqrt{11.2} = 3.35 \text{ m/s}$$

Rotational velocity:

$$\omega = \frac{v}{R} = \frac{3.35}{0.050} = 67 \text{ rad/s}$$

Distractors:

- a) 3.3 is v (m/s), not ω.
- b) 33 divides by the diameter (0.10 m) instead of the radius.
- c) 56 uses the ring formula, $\sqrt{gh}$.
- d) 65 uses the disk formula, $\sqrt{\frac{4}{3}gh}$.

14. Person runs and jumps on inside the rim — c) 0.98 rad/s

Source: Ch 11 notes p. 4/8; 2015 Exam 2 #12. **Notecard:** Angular momentum.

Notecard with $+mvr$ (running with the rotation) and $I_{add} = mr^2$, where $r = 1.5$ m is where the child lands:

$$\omega_f = \frac{I_1\omega_1 + mvr}{I_1 + mr^2}$$

Platform: $I_1 = \frac{1}{2}(200)(2.5)^2 = 625 \text{ kg} \cdot \text{m}^2$.

Plug in:

$$\omega_f = \frac{(625)(0.80) + (30)(4.0)(1.5)}{625 + (30)(1.5)^2}$$

$$= \frac{500 + 180}{625 + 67.5} = \frac{680}{692.5} = 0.98 \text{ rad/s}$$

Use $r = 1.5$ m in **both** places (the child's $L = mvr$ and their $I = mr^2$), not the rim radius.

Distractors:

- a) 0.72 ignores the child's running momentum.
- b) 0.80 is no change.
- d) 1.09 forgets to add the child's mr^2.
- e) 1.16 uses the rim radius 2.5 m in mvr.

15. Rotational kinematics, time from an angle — a) 5.0 s

Source: Ch 10 notes p. 1/8. **Notecard:** Rot. kinematics.

ω_f isn't given or asked for, so use the equation without it:

$$\Delta\theta = \omega_i t + \tfrac{1}{2}\alpha t^2$$

Solve for t (starting from rest, $\omega_i = 0$):

$$\Delta\theta = \tfrac{1}{2}\alpha t^2$$

$$t = \sqrt{\frac{2\,\Delta\theta}{\alpha}}$$

Plug in:

$$t = \sqrt{\frac{2(50)}{4.0}} = \sqrt{25} = 5.0 \text{ s}$$

Distractors:

- b) 6.25 is $\Delta\theta/2\alpha$ with no root.
- c) 12.5 is $\Delta\theta/\alpha$.
- d) 25 is $2\Delta\theta/\alpha$ (forgot the square root).
- e) 50 is just $\Delta\theta$.